EXPERIMENT-5

5. Scenario: You are a data analyst working for a car manufacturing company. As part of your analysis, you have a dataset containing information about the fuel efficiency of different car models. The dataset is stored in a NumPy array named fuel_efficiency, where each element represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models.

Question: How would you use NumPy arrays and arithmetic operations to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models?

Aim

To calculate the average fuel efficiency of different car models and determine the percentage improvement between two car models using NumPy.

Algorithm

1. Import the NumPy library.
2. Create a NumPy array containing the fuel efficiency of different car models.
3. Calculate the average fuel efficiency using np.mean().
4. Select the fuel efficiency of two car models.
5. Calculate the percentage improvement using the formula:
   Percentage Improvement = ((New − Old) / Old) × 100
6. Display the average fuel efficiency and percentage improvement.

Code

```
import numpy as np

fuel_efficiency = np.array([25, 30, 28, 35, 32])

average_efficiency = np.mean(fuel_efficiency)
```

```
model1 = fuel_efficiency[0]

model2 = fuel_efficiency[1]

percentage_improvement = ((model2 - model1) / model1) * 100

print("Average fuel efficiency:",

    average_efficiency, "MPG")


print("Percentage improvement from Model 1 to Model 2:",

    percentage_improvement, "%")
```

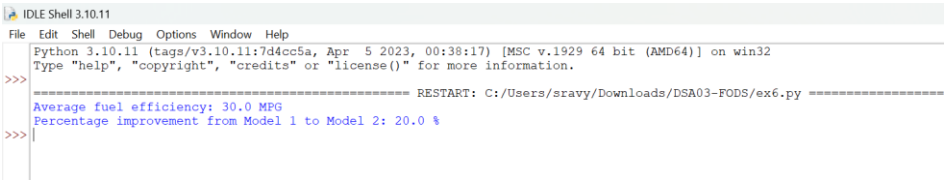
Input

25, 30, 28, 35, 32 MPG

Output

Average fuel efficiency: 30.0 MPG

Percentage improvement from Model 1 to Model 2: 20.0 %



Result

Thus, the average fuel efficiency was calculated successfully as **30 MPG**, and the percentage improvement from Model 1 to Model 2 was **20%**.